

GLOSSARIES

VARIABLE – CHARACTERISTICS WHICH CAN TAKE DIFFERENT VALUES

ATTRIBUTE—CATEGORICAL VARIABLE WITH DISTINCT PROPERTIES (*discussed in later sections*)

MARGINAL FREQUENCY—TOTAL FREQUENCY FOR ONE ATTRIBUTE, IGNORING THE CLASSIFICATION BY THE OTHER ATTRIBUTE/ATTRIBUTES.

1.1 CONCEPTS OF ASSOCIATION AND INDEPENDENCE

Mama reports to Physician Chhanhima that he smokes and feels not well about his health. Now based on the following 3 dataset we may estimate the decision of Dr. Chhanhima.

Dataset A: N=1000 (Distribution based upon 1000 People's Observation)

	Lung Disease (D)	No Lung Disease ($\bar{D}$)
Smoker (S)	4	196
Non-Smoker ($\bar{S}$)	16	784

Dataset B: N=1000 (Distribution based upon 1000 People's Observation)

	Lung Disease (D)	No Lung Disease ($\bar{D}$)
Smoker (S)	100	100
Non-Smoker ($\bar{S}$)	8	792

Dataset C: N=1000 (Distribution based upon 1000 People's Observation)

	Lung Disease (D)	No Lung Disease ($\bar{D}$)
Smoker (S)	2	198
Non-Smoker ($\bar{S}$)	100	700

If Dataset A is the reality of Dr. Chhanhima's world then we estimate nothing (**As Smoking & Disease, are independence of each other**). And if Dataset B is reality, we estimate of Disease (**Smoking & Disease are in association with each other**). And if By Chance, dataset C is reality, we estimate of No Disease.

For Distribution A:

$$P(D/S) = P(D) = 0.02 = P(D/\bar{S})$$

Equivalently

$$P(D \cap S) = P(D)P(S)$$

Knowing Smoking doesn't change our estimation of disease.

The relationship of Independence between two variables/attributes is when One attribute doesn't give any Predictive Information or doesn't change the Estimation of the other attribute in study. The two attributes are independent of each other. Occurrence of one characteristic doesn't affect the occurrence of another characteristic. Their expected joint frequencies from the separate marginal frequencies equal with the respective observed joint frequencies.

For Distribution B:

$$P(D/S) = 0.5 > P(D) = 0.108 > P(D/\bar{S}) = 0.01$$

Equivalently

$$P(D \cap S) \neq P(D)P(S)$$

Knowing Smoking change our estimation of disease in positive direction, i.e., that is we may estimate a higher chance of occurrence of disease.

The relationship of Positive Association between two attributes is when One attribute changes the estimation of another attribute in positive direction, that is occurrence of one characteristic make the occurrence of another characteristic more probable. Their expected joint frequencies obtained from the separate marginal frequencies doesn't equal with the respective observed joint frequencies.

For Distribution C:

$$P(D/S) = 0.01 < P(D) = 0.102 < P(D/\bar{S}) = 0.125$$

Equivalently

$$P(D \cap S) \neq P(D)P(S)$$

Knowing Smoking change our estimation of disease in negative direction, i.e., that is we may estimate a lower chance of occurrence of disease.

The relationship of Negative Association between two attributes is when One attribute changes the estimation of another attribute in negative direction, that is occurrence of one characteristic make the occurrence of another characteristic less probable. Their expected joint frequencies obtained from the separate marginal frequencies doesn't equal with the respective observed joint frequencies.

So, in broad, Association is a relationship between two variables in study in which occurrence of one attribute affects the occurrence of another

attribute in study. And mathematically, $P(X \cap Y) \neq P(X)P(Y)$ for any two attributes X and Y under study.

1.2 ATTRIBUTE: WHAT AND WHY

In statistics, Attributes are categorical or qualitative characteristics of a population or a statistical unit which can't be measure in numerical terms. Their presence and absence can be determined and the units possessing and not possessing the characteristics or attributes can be classified into distinct groups for further mathematical and statistical analysis.

E.g., Employment, Literacy, Loan Default, Insurance, Attendance, Education.

Confusion Point: In classical Theory of Attributes, the basic unit is a dichotomous attribute, i.e., the simple attribute will have only two classes. Absence of the attribute or presence of the attribute. However Qualitative Variable is different, For Example, Education as a qualitative variable may have values as no education, primary education, secondary education, higher education, doctorate etc. etc. But Education as an attribute can only have Having Education (Presence of the attribute) and not having Education (Absence of the attribute).

A class is a group of statistical units having the same attribute status. A class frequency is the number of statistical units belonging to that particular class. For Example, all those people who have employment will form a group that is a class so for all those not having employment. And the number of each class is a class frequency. Let Notation of Employment be E for attribute Employment, it could have two different class only as either a person (statistical unit) is employed or unemployed, so there is presence of E, E or absence of E, $\bar{E}$.

The class formed by presence of the attribute is known as Positive Class of the Attribute and the frequency of the positive class is known as Positive Class frequency of the attribute. Let say the attribute is denoted as A.

Notations: Positive Class— A

Positive Class Frequency— (A) of the specific attribute.

The class formed by absence of the attribute is known as Negative Class of the Attribute and the frequency of the Negative class is known as Negative Class frequency of the attribute. Let say the attribute is denoted as A.

Notations: Negative Class— $\bar{A}$

Negative Class Frequency— $(\bar{A})$ of the specific attribute.

Based on Attributes/characteristics Considered Number of classes and type of Classes will vary

Classification of attributes based on Classes they will possess

Simple Attribute: Only one Characteristic/attribute of a statistical unit is considered. E.g. Literacy L, Education E, two classes Positive and Negative.

Manifold Attribute: Combination of Two or More dichotomous characteristics or attributes is considered simultaneously.

Now consider two attributes E and L,

A person will exactly be in EL or $\bar{E}L$ or $E\bar{L}$ or $\bar{E}\bar{L}$ classes. A statistical unit (here person) will be exactly in only one class, can't be in more than one. All classes are mutually exclusive. Such classes are known as **Ultimate Classes** of the Manifold or Joint Attribute. And the frequency of such mutually exclusive classes (Ultimate Classes) are called **Ultimate Class frequencies**. In Ultimate Class, When all attributes are present, we may call it positive class and when all are absent, we may call it negative class, rest are mix class.

Number of Ultimate classes = 2^k for k dichotomous attributes.

Number of total observations = *Summation of Ultimate Class Frequencies*

For the above example of 2 attributes, $N = (EL) + (\bar{E}L) + (E\bar{L}) + (\bar{E}\bar{L})$

First Order Class: In a Manifold Attribute ignoring all the other attributes considering only one attribute let say A, the classes we obtain so is known as First Order Class. And the frequencies so obtain from these First Order Class is known as **First Order Class Frequency**. It is same with Positive and Negative Class of simple attribute, just taken from a joint attribute, one attribute in consideration.

Relations between **First Order and Ultimate Class Frequencies** of Manifold Attribute of 2 attributes.

$$(A) = (AB) + (A\bar{B})$$

$$(\bar{A}) = (\bar{A}B) + (\bar{A}\bar{B})$$

$$N = (A) + (\bar{A})$$

Same goes for attribute B.

1.3 Classification/Contingency Table

How the Observations of Statistical Units are distributed as per joint Characteristics can be tabulated for better inferences and that table is known as Contingency Table.

Let Disease D and Smoking S be two attributes, the 2X2 contingency table is tabulated Below.

	Having Disease D	Not having Disease $\bar{D}$	<i>Total</i>
Smoker S	(SD)	$(S\bar{D})$	(S)
Not Smoker $\bar{S}$	$(\bar{S}D)$	$(\bar{S}\bar{D})$	$(\bar{S})$
<i>Total</i>	(D)	$(\bar{D})$	<i>N Value</i>

Benefits of Contingency Table and Determination of Association from the Table:

a) Finding the Missing Frequencies

1. The center 4 frequencies are Ultimate Class Frequencies (UCF). Addition of UCF column wise or row wise will give the first order class frequencies.
2. Column or Row of those First Order Class Frequencies addition will give total number of Observation
3. That's how we can make several formulas and complete the table in case of missing frequencies.

b) Determination of Association and Independence

Expected Frequencies: $E_{ij} = \frac{(Row\ total_i)(Column\ total_j)}{N}$

Example...

For the SD cell $E[(SD)] = \frac{(S)(D)}{N}$

For the $S\bar{D}$ cell $E[(S\bar{D})] = \frac{S\bar{D}}{N}$

For Independence, the observed frequencies should equal with the expected frequencies.

For Association, the observed frequencies won't be equal with the expected frequencies.

c) Finding Conditional Probability

Joint Frequencies (UCF) and Marginal Frequencies (First Order Frequency) side by side helps in finding the Conditional Probability Easily. E.g.,

$$P(D/S) = \frac{(SD)}{(S)}$$

Let say S is the given attribute and D is the attribute being estimated i.e., we want to estimate D from S (Having Disease from Smoking)

The Conditional Probability Helps in finding Association or Independence.

For Independence: $P(D \cap S) = P(S)P(D)$

For Association: $P(D \cap S) \neq P(S)P(D)$

Positive association: $P(D/S) > P(D/\bar{S})$

Negative Association: $P(D/S) < P(D/\bar{S})$

Same for $\bar{D}$, $\bar{S}$ and other relations.

2. Conditions for two Attributes to be Independent in terms of their Frequencies:

Let below be the contingency Table for two Attributes A and B:

	B	$\bar{B}$	$Total$
A	(AB)	$(A\bar{B})$	(A)
$\bar{A}$	$(\bar{A}B)$	$(\bar{A}\bar{B})$	$(\bar{A})$
$Total$	(B)	$(\bar{B})$	$N \text{ Value}$

Under Independence,
Expected Frequencies,

$$E(AB) = \frac{(A)(B)}{N} = (AB)$$

$$(A\bar{B}) = \frac{(A)(\bar{B})}{N}$$

$$(\bar{A}B) = \frac{(\bar{A})(B)}{N}$$

$$(\bar{A}\bar{B}) = \frac{(\bar{A})(\bar{B})}{N}$$

$$(AB)(\bar{A}\bar{B}) = (A\bar{B})(\bar{A}B)$$

When these frequencies Equations are holds true, it means the two Attributes A and B are in Independence Relationship. And their Expected Frequencies Equals their observed frequencies.

3. An Example of Attributes and Classification Table with numbers

Total Students, $N=100$, Let Mathematics is denoted by M and Statistics is be denoted by S. And given

$$(M) = 50 \text{ and } (\bar{M}) = N - (M) = 50$$

$$(S) = 60 \text{ and } (\bar{S}) = N - (S) = 40$$

$$(SM) = 30$$

Now, let find the missing values,

$$(S) = (SM) + (S\bar{M}), \quad (S\bar{M}) = 30$$

$$(M) = (SM) + (\bar{S}M), \quad (\bar{S}M) = 20$$

$$(\bar{S}) = (\bar{S}M) + (\bar{S}\bar{M}), \quad (\bar{S}\bar{M}) = 20$$

The 2-way classification table is tabulated below

	M	$\bar{M}$	$Total$
S	30	30	60
$\bar{S}$	20	20	40
$Total$	50	50	100

Determination of Association and Independence

The expected joint frequencies from separate Marginal Frequencies

$$E(SM) = \frac{(S)(M)}{N} = 30 = (SM), \text{Original observed Frequency}$$

$$E(S\bar{M}) = \frac{(S)(\bar{M})}{N} = 30 = (S\bar{M})$$

$$E(\bar{S}M) = \frac{(\bar{S})(M)}{N} = 20 = (\bar{S}M)$$

$$\frac{(\bar{S})(\bar{M})}{N} = 20 = (\bar{S}\bar{M})$$

And moreover, $(SM)(\bar{S}\bar{M}) = 600 = (S\bar{M})(\bar{S}M)$

As all the expected frequencies are equal with the observed frequencies, therefore Passing in Mathematics M and Statistics S are completely Independent of each other.

REFERENCES

1. Miss Hruaii Notes
2. Penn University Statistics Course Material